Basic Inequalities, Cauchy, Triangle Inequality, and Coordinate Geometry

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§1.1 Basic inequalities

The first page is a compact inequality note. It begins with the elementary observation that squares are nonnegative. For real numbers,

$$(a-b)^2 \geq 0,$$

so

$$a^2 + b^2 \geq 2ab,$$

with equality iff $a = b$. For three variables,

$$a^2 + b^2 + c^2 \geq ab + bc + ca,$$

because

$$2(a^2 + b^2 + c^2 - ab - bc - ca) = (a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0.$$

The handwritten page marks equality conditions carefully: equality occurs when all variables involved in the squared differences coincide.

§1.2 AM–GM

For $a, b \geq 0$,

$$\frac{a+b}{2} \geq \sqrt{ab},$$

with equality iff $a = b$. Equivalently,

$$a + b \geq 2\sqrt{ab}.$$

The note also records the three-variable form

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}, \qquad a, b, c \geq 0,$$

and uses it repeatedly in examples.

A typical application is to expressions such as

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc} \quad \Longrightarrow \quad a + b + c \geq 3\sqrt[3]{abc}.$$

The useful habit is to look for terms whose product is fixed. AM–GM then turns a fixed product into a lower bound for a sum.

§1.3 Cauchy's inequality

The note states Cauchy's inequality in two forms. The vector form is

$$(a_1b_1 + \cdots + a_nb_n)^2 \le (a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2).$$

The Engel or Titu Andreescu form is

$$\frac{x_1^2}{a_1} + \cdots + \frac{x_n^2}{a_n} \ge \frac{(x_1 + \cdots + x_n)^2}{a_1 + \cdots + a_n}, \qquad a_i > 0.$$

This second form is often what the handwritten examples use. It is obtained by applying Cauchy to

$$\left(\sum_i \frac{x_i^2}{a_i}\right)\left(\sum_i a_i\right) \ge \left(\sum_i x_i\right)^2.$$

§1.4 Triangle inequality and its variants

The page then discusses absolute values:

$$|a+b| \le |a| + |b|, \qquad |a-b| \le |a-c| + |c-b|.$$

The second formula is the triangle inequality with an intermediate point c. The note also uses the reverse triangle inequality

$$||a| - |b|| \le |a-b|.$$

A useful proof is

$$|a| = |a-b+b| \le |a-b| + |b|, \qquad |b| = |b-a+a| \le |a-b| + |a|.$$

Together these imply the reverse inequality.

The handwritten warning is that absolute value inequalities are not merely algebraic signs. They encode distances on the real line.

§1.5 Worked AM–GM examples

One example asks for a lower bound of expressions of the form

$$x + \frac{1}{x}, \qquad x > 0.$$

By AM–GM,

$$x + \frac{1}{x} \ge 2.$$

Similarly,

$$x + y + z \ge 3\sqrt[3]{xyz}$$

when $x, y, z > 0$. If a problem fixes xyz, this gives a direct lower bound for the sum.

Another common pattern is

$$\frac{x}{y} + \frac{y}{z} + \frac{z}{x} \ge 3, \qquad x, y, z > 0,$$

because the product of the three terms is 1.

§1.6 A Cauchy example

For positive a, b, c, a typical inequality is

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{(a+b+c)^2}{2(a+b+c)} = \frac{a+b+c}{2}.$$

This is exactly Engel form of Cauchy. The handwritten page uses this kind of manipulation to turn a sum of fractions into one clean square divided by one clean denominator.

§1.7 Coordinate geometry at the end of the note

The last page shifts to coordinate geometry. The diagrams show points, vectors, and lines, with formulas for distances and slopes. The essential formulas are:

$$AB = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2},$$

$$m = \frac{y_2 - y_1}{x_2 - x_1},$$

and the point-slope form of a line

$$y - y_0 = m(x - x_0).$$

A recurring idea is that geometric claims can be turned into algebraic equalities. For example, perpendicularity of two nonvertical lines is

$$m_1 m_2 = -1,$$

while parallelism is

$$m_1 = m_2.$$

Distance and slope are the coordinate versions of length and direction.

§1.8 What the examples reveal: three structures

This note is about choosing the right inequality language. Squares prove elementary inequalities, AM–GM handles fixed products, Cauchy handles dot products and sums of fractions, and triangle inequalities handle distances. The advanced version of the same habit is to identify the structure behind the inequality before calculating.

Elementary inequality	Higher structure	Equality meaning
Cauchy–Schwarz	inner product / projection / Gram matrix	vectors are dependent
Triangle inequality	normed vector space	same direction in strictly convex cases
AM–GM	convexity or concavity of log	all variables equal
Positive square	positive semidefinite cone	quadratic form degenerates

§1.9 Cauchy–Schwarz and Gram determinants

Cauchy–Schwarz says
$$|\langle x, y\rangle| \le \|x\|\|y\|.$$
It follows from the nonnegativity of
$$\|x - ty\|^2 \ge 0$$
or, equivalently, from positivity of the Gram matrix
$$G(x, y) = \begin{pmatrix} \langle x, x\rangle & \langle x, y\rangle \\ \langle y, x\rangle & \langle y, y\rangle \end{pmatrix}.$$
The inequality $\det G(x, y) \ge 0$ is exactly Cauchy–Schwarz. Equality means G has rank one, i.e. the two vectors are linearly dependent.

In Hilbert spaces, the same inequality controls Fourier coefficients:
$$|\langle f, e_n\rangle| \le \|f\|_2 \|e_n\|_2.$$
This is why an inequality first met in finite sums becomes indispensable in functional analysis.

§1.10 Triangle inequality and normed spaces

The triangle inequality
$$\|x + y\| \le \|x\| + \|y\|$$
defines the geometry of a normed vector space. The usual absolute value, Euclidean length, sup norm, and L^p norms are all instances. Minkowski's inequality is precisely the triangle inequality in L^p:
$$\|f + g\|_p \le \|f\|_p + \|g\|_p, \qquad \|f\|_p = \left(\int |f|^p\right)^{1/p}.$$
Thus a contest-level inequality about square roots and sums is structurally the same statement as the triangle inequality for functions.

§1.11 AM–GM, convexity, and cones

For positive variables, AM–GM follows from concavity of log:
$$\log\left(\frac{1}{n}\sum_i x_i\right) \ge \frac{1}{n}\sum_i \log x_i.$$
The equality condition says all variables are equal, meaning there is no spread.

Many inequalities also describe membership in a convex cone. For instance,
$$A \succeq 0 \quad \Longleftrightarrow \quad x^T A x \ge 0 \text{ for all } x$$
defines the cone of positive semidefinite matrices. Basic "square is nonnegative" arguments are the one-dimensional shadow of this cone viewpoint.